

$(u_1, v_1)(u_2, v_2)$ such that either $u_1u_2 \in E(G)$ and $v_1 = v_2$, or $v_1v_2 \in E(H)$ and $u_1 = u_2$, or $u_1u_2 \in E(G)$ and $v_1v_2 \in E(H)$. By definition, the graph $G \square H$ is the spanning subgraph of the graph $G \boxtimes H$. By using the definition and structure of Cartesian product, Li and Sun derived the following.

Theorem 2.55 [46] *Let $G^* = G_1 \square G_2 \square \cdots \square G_k$ ($k \geq 2$), where each G_i ($1 \leq i \leq k$) is connected. Then we have*

$$rc(G^*) \leq \sum_{i=1}^k rc(G_i).$$

Moreover, if $diam(G_i) = rc(G_i)$ for each G_i , then the equality holds.

We know that there are infinity many graph with $diam(G) = rc(G)$, such as the unit interval graphs shown in Theorem 2.41. The following problem could be interesting but maybe difficult:

Problem 2.56 *Characterize the graphs G with $rc(G) = diam(G)$, or give some sufficient conditions to guarantee that $rc(G) = diam(G)$.*

Similar problem for $src(G)$ can be considered.

Let $\overline{G^*} = G_1 \boxtimes G_2 \boxtimes \cdots \boxtimes G_k$ ($k \geq 2$), where each G_i ($1 \leq i \leq k$) is connected. Since the Cartesian product of any two graphs is a spanning subgraph of their strong product, G^* is the spanning subgraph of $\overline{G^*}$, then we have the following result.

Corollary 2.57 [46] *Let $\overline{G^*} = G_1 \boxtimes G_2 \boxtimes \cdots \boxtimes G_k$ ($k \geq 2$), where each G_i ($1 \leq i \leq k$) is connected. Then we have*

$$rc(\overline{G^*}) \leq \sum_{i=1}^k rc(G_i).$$

For $i = 1, 2, \dots, r$, let $m_i \geq 2$ be given integers. Consider the graph G whose vertices are the r -tuples $b_1b_2 \cdots b_r$ with $b_i \in \{0, 1, \dots, m_i - 1\}$, and let two vertices be adjacent if the corresponding tuples differ in precisely one place. Such a graph is called a *Hamming graph*. Clearly, a graph G is a Hamming graph if and only if it can be written in the form $G = K_{m_1} \square K_{m_2} \cdots \square K_{m_r}$ for some $r \geq 1$, where $m_i \geq 2$ for all i . So we call G a Hamming graph with r factors. A Hamming graph is a *hypercube* (or *r -cube*) [28], denoted by Q_r , if and only $m_i = 2$ for all i . The concept of Hamming graph is useful in communication networks [32].

Corollary 2.58 [46] *If G is a Hamming graph with r factors, then $rc(G) = r$. In particular, $rc(Q_r) = r$.*

The *composition* (lexicographic product) of two graphs G and H is the simple graph $G[H]$ with vertex set $V(G) \times V(H)$ in which (u, v) is adjacent to (u', v') if and only if either

$uu' \in E(G)$ or $u = u'$ and $vv' \in E(H)$. By definition, $G[H]$ can be obtained from G by substituting a copy H_v of H for every vertex v of G and by joining all vertices of H_v with all vertices of H_u if $uv \in E(G)$. Note that $G[H]$ is connected if and only if G is connected. By definition, it is easy to show: If G is complete, then $\text{diam}(G[H]) = 1$ if H is complete (as now $G[H]$ is complete), $\text{diam}(G[H]) = 2$ if H is not complete; If G is not complete, then $\text{diam}(G[H]) = \text{diam}(G)$. Then we have the following result.

Theorem 2.59 [46] *If G and H are two graphs and G is connected, then we have*

(1) *if H is complete, then*

$$rc(G[H]) \leq rc(G).$$

In particular, if $\text{diam}(G) = rc(G)$, then $rc(G[H]) = rc(G)$.

(2) *if H is not complete, then*

$$rc(G[H]) \leq rc(G) + 1.$$

In particular, if $\text{diam}(G) = rc(G)$, then $\text{diam}(G[H]) = 2$ if G is complete and $rc(G) \leq \text{diam}(G[H]) \leq rc(G) + 1$ if G is not complete.

In [46], Li and Sun also investigated other graph operations, such as the union of graphs which we will not introduce here.

2.6 An upper bound for strong rainbow connection number

The topic of rainbow connection number is fairly interesting and recently a series papers have been published about it. The strong rainbow connection number is also interesting, and by definition, the investigation of it is more challenging than that of the rainbow connection number. However, there are very few papers that have been published about it. In [12], Chartrand, Johns, McKeon and Zhang determined the precise strong rainbow connection numbers for some special graph classes including trees, complete graphs, wheels and complete bipartite (multipartite) graphs as shown in Subsection 2.1. However, for a general graph G , it is almost impossible to give the precise value for $\text{src}(G)$, so we aim to give upper bounds for it according to some graph parameters. Li and Sun [47] derived a sharp upper bound for $\text{src}(G)$ according to the number of edge-disjoint triangles (if exist) in a graph G , and give a necessary and sufficient condition for the sharpness. We need to introduce a new graph class.

Recall that a *block* of a connected graph G is a maximal connected subgraph without a cut vertex. Thus, every block of G is either a maximal 2-connected subgraph or a bridge. We now introduce a new graph class. For a connected graph G , we say $G \in \overline{\mathcal{G}}_t$, if it satisfies the following conditions:

C_1 . Each block of G is a bridge or a triangle;

C_2 . G contains exactly t triangles;

C_3 . Each triangle contains at least one vertex of degree two in G .

By the definition, each graph $G \in \overline{\mathcal{G}}_t$ is formed by (edge-disjoint) triangles and paths (may be trivial), these triangles and paths fit together in a treelike structure, and G contains no cycles but the t (edge-disjoint) triangles. For example, see Figure 2.4, here $t = 2$, and

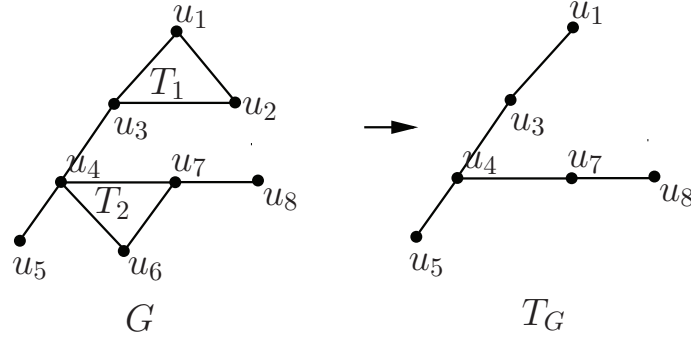


Figure 2.4 An example of $G \in \overline{\mathcal{G}}_t$ with $t = 2$.

u_1 , u_2 and u_6 are vertices of degree 2 in G . If a tree is obtained from a graph $G \in \overline{\mathcal{G}}_t$ by deleting one vertex of degree 2 from each triangle, then we call this tree is a D_2 -tree of G , denoted by T_G . For example, in Figure 2.4, T_G is a D_2 -tree of G . Clearly, the D_2 -tree is not unique, since in this example, we can obtain another D_2 -tree by deleting the vertex u_1 instead of u_2 . On the other hand, we can say that any element of $\overline{\mathcal{G}}_t$ can be obtained from a tree by adding t new vertices of degree 2. It is easy to show that the number of edges of T_G is $m - 2t$ where m is the number of edges of G .

They derived the following result.

Theorem 2.60 [47] *If G is a graph with m edges and t edge-disjoint triangles, then*

$$src(G) \leq m - 2t,$$

the equality holds if and only if $G \in \overline{\mathcal{G}}_t$.

In [1], Ahadi and Dehghan also derived an upper bound for strongly regular graph: if G is an $SRG(n, r, \lambda, \mu)$, then $src(G) \leq \lceil (e(4\mu r - 4\mu\lambda - 6\mu + 1))^{\frac{1}{\mu}} \rceil$. However, we do not know whether it is sharp.

Unlike rainbow connection number, which is a monotone graph property (adding edges never increases the rainbow connection number), this is not the case for the strong rainbow connection number (see Figure 2.1 for an example). The investigation of strong rainbow connection number is much harder than that of rainbow connection number. Chakraborty, Fischer, Matsliah and Yuster gave the following conjecture.

Conjecture 2.61 [10] *If G is a connected graph with minimum degree at least en , then it has a bounded strong rainbow connection number.*